Tampere University

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WHICH DEVELOPMENTS LED TO THE FOURTH INDUSTRIAL REVOLUTION

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ABSTRACT

Leevi Kukkonen: The developments which led to the fourth industrial revolution
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This thesis aims to observe and analyse different technological developments which led to the fourth industrial revolution, also known as industry 4.0. The current industrial revolution builds heavily upon the third industrial revolution's innovations, but there does still exist a need to analyse all of the previous revolutions in order to find common patterns.

While the second and first industrial revolutions focused more on the transition from using human muscles as a power source in labour to supplementing that with ever more powerful machinery, powered by new and abundant power sources such, the third brought on a digital revolution where machine to machine communication was made ever more accessible. The third industrial revolution spawned things such as the internet, personal computers and mobile devices, which the fourth industrial revolution built upon. The fourth industrial revolution is thus characterized by the interconnectivity of machines and the supplementation of brain power with powerful data gathering, processing and analysis tools.

The key innovations of the fourth industrial revolution are thusly related heavily to on demand interconnectivity, which is why the internet of things is such an important development within this revolution. Some other notable technologies include 3D-printing, cloud computing, big data and artificial intelligence. These technologies play off one another and form complex systems that require less and less human involvement.

When looking at the data from industrial revolutions, it becomes abundantly clear that there are no clear lines when a revolution begins and ends, which is why finding metrics to measure these revolutions is so difficult. Industrial revolutions share some characteristics, but their effects are not easily observable, and because innovation and development doesn't cease between revolutions, the term becomes more of a commonly accepted term for a period of time when the nature of work shifts noticeably.

Ultimately, the fourth industrial revolution is too early in its infancy for any real analysis on what specific technologies caused it and whether or not it actually is a brand-new revolution or just an extension of the third one. This does not however mean that the technologies that the current literature agrees are developments of the fourth industrial revolution aren't shaping the nature of work significantly. The technologies presented on this thesis are garnering a noticeable amount of attention and have undoubtedly had an effect on industry 4.0.

Keywords: The fourth industrial revolution, internet of things, industry 4.0, IoT

The originality of this thesis has been checked using the Turnitin Originality Check service.

TIIVISTELMÄ

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Tämän tutkimuksen tavoitteena oli selvittää ja analysoida niitä teknologisia kehitysaskeleita, jotka johtivat teollisuus 4.0:na tunnettuun neljänteen teolliseen vallankumoukseen. Työ perustui laajaan kirjallisuuskatsaukseen ja teoreettiseen pohdintaan, jossa tarkasteltiin kolmen aiemman teollisen vallankumouksen innovaatioita ja niiden vaikutuksia työn luonteeseen, kun innovaatioita alettiin ensimmäistä kertaa hyödyntää laajemmin. Tutkimuskysymykset keskittyivät erityisesti siihen, mitä yhtäläisyyksiä voidaan löytää teollisten vallankumousten väliltä, ja ovatko nykyisen neljännen teollisten vallankumousten avainteknologiat saaneet aikaan samanlaisia vaikutuksia, kuin edellisten teollisten vallankumousten merkittävimmät innovaatiot.

Tutkimuksessa hyödynnettiin sekä historiallista että nykyaikaista tietoa teollisten vallankumousten aikana tapahtuneista muutoksista työn laatuun, sekä tuottavuuteen. Neljännen teollisen vallankumouksen ominaispiirteitä ovat laitteiden väliset yhteydet, sekä ajattelutyön tukeminen, ja näihin liittyvät tarkasteltavat keskeiset teknologiat olivat, esineiden internet (IoT), 3D-tulostus, pilvipalvelut, suurten tietomassojen käsittelyn, sekä tekoäly.

Keskeiset tulokset osoittivat, että neljännen teollisen vallankumouksen teknologiat tukevat toisiaan luoden monimutkaisia järjestelmiä, jotka vähentävät ihmisen osallistumista tuotantoprosesseihin. Tutkimuksen perusteella ei voida suoranaisesti kuitenkaan sanoa, että onko neljäs teollinen vallankumous oma ilmiönsä, vai vain jatkoa kolmannelle teolliselle vallankumoukselle.

Teollisten vallankumousten välille ei löytynyt tarkkoja mittaristoja, joten työn laadun muuttuminen oli lopulta ainoa kaikkia teollisia vallankumouksia yhdistävä tekijä. Työn päätelmissä korostettiin, että vaikka neljäs teollinen vallankumous on vielä meneillään ja sen täydet vaikutukset ovat tuntemattomia, sen mukanaan tuomat teknologiat muovaavat jo nyt työn luonnetta globaalisti. Lopuksi todettiin, että tarkempien analyysien tekeminen neljännen vallankumouksen vaikutuksista edellyttää nykyisten työn luonteeseen tapahtuneiden muutosten tutkimista, kun neljännestä teollisesta vallankumouksen alkamisesta on kulunut useita vuosia, tai jopa vuosikymmeniä.

Avainsanat: Neljäs teollinen vallankumous, Internet of things, Esineiden internet, Teollisuus 4.0, IoT

USE OF AI IN THESIS

I have utilised AI tools in my thesis:

☐ No
☒ Yes

The AI tools utilised in my thesis and their purposes are described below:

Names and versions of AI tools: OpenAI's ChatGPT-4

Purpose of using AI tools: Testing out the capabilities of the most widely used AI tool for consumers. I tried generating simple functions for Python, generating images and inputting large text files and having this AI tool analyse them. Nothing that was generated was used in this work. The testing was done due to AI being one of the key technologies discussed in this thesis.

Sections where AI tools were used: Chapter 2.3.5. No text for this chapter was generated, but some parts were written based on having tried these AI tools myself.

I acknowledge that I am fully responsible for the entire content of my thesis, including the parts generated by AI, and accept accountability for any violations of ethical standards in publications.

ALKUSANAT

Tämä oli hieman työn takana. Suuret ja lämpimät kiitokset teille jotka kannustitte, sekä teille jotka painostitte, työ ei olisi ilman molempia osapuolia tullut valmiiksi.

Tampereella, 31.10.2024

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1. INTRODUCTION

The fourth industrial revolution, also known as the industrial internet of things, or industry 4.0 is latest of industry shaking periods, when various key technologies alter the fundamental nature of work and lead to vast changes in the technological, social and economic landscapes. The fourth industrial revolution is an extremely new development, only being first used in academic circles less than 10 years ago (Schwab 2016), and as such the effects it has are only still developing and the full-scale picture will only become clear after a few decades.

The fourth industrial revolution and its key technologies share some similarities with its predecessors in the way they are transforming labour. By finding these similarities and patterns in how past industrial revolutions have played out, the goal is to nail down whether these innovations have had a similar effect to those of the past revolutions. Difficulties will undoubtedly occur when trying to find patterns in incomplete data, as the fourth industrial revolution has by any definition, lasted only a fraction of the time the previous ones have.

The aim of this thesis is to observe what can be learned from previous industrial revolutions and how can that knowledge be contrasted with the current fourth one. The key technologies of the fourth industrial revolution are presented and by analysing data from the past 250 years of industrial revolutions, hopefully an answer to the questions “What makes an industrial revolution” and “Did these new technologies spawn a new industrial revolution” can be answered.

2. CHARACTERISTICS OF INDUSTRIAL REVOLUTIONS

2.1 From the first to the third industrial revolution

To note, industrial revolutions don't occur simultaneously all over the world, but rather different regions experience them at different times. For the rest of this thesis, when referring to a specific industrial revolution, it can be read as meaning the industrial revolution which occurred in the current western world, which has historically experienced industrial revolutions first, and roughly at the same time.

Traditionally the term industrial revolution is used to describe the period during which industrial output substantially increased, productivity grew, and the affected area saw a spike in innovation and some rather rapid social reforms (Hudson 2009). This first occurred in England during the late 18th and early 19th centuries, where the conditions were ripe for the empire to start producing and exporting a large number of cheaply made products all over the world. The first industrial revolution can be considered a turning point in history, but it was far from the last. Some of the key technologies introduced during this time were the steam engine, advanced textile manufacturing processes, and telegraph communication.

To a somewhat diminished fanfare, the second industrial revolution occurred around a hundred years after the start of the first one. Innovation never ceased between the two revolutions, far from it, but the second industrial revolution saw the widespread adoption of things such as electricity, internal combustion engines, advances in chemicals such as petroleum and new communication technologies such as the radio. Paradoxically, the expected uptick in productivity happened slowly during this revolution, but it did nevertheless occur (Atkeson et al. 2001). This period is often referred to as the gilded age, and while the term refers more to the economic changes occurring at the time, it can be argued that they are just an extension of the wider industrial revolution.

Three occurrences hardly form a trend, but it is interesting to note that the third industrial revolution once again started around a hundred years after the start of the second one. This time, the industrial revolution was brought on mostly by advances in information and communication technologies and their widespread adoption. Most people alive today have lived through at least a part of this revolution, and for the average consumer, mobile phones, PC's and the internet are the most visible outcomes of this revolution. These were made possible by one of the most important innovations of the modern age, the transistor, which allowed processing units to work fully digitally. With information age devices becoming as readily accessible as they are, the third industrial revolution saw a vast change in day-to-day communication. As of October 2023, more than 65% of the world had internet access (Petrosyan 2023) and the UN considers disrupting internet access as a violation of human rights (Vincent 2016).

2.2 From the third industrial revolution to the fourth

There is some debate about when exactly the third industrial revolution began, but generally it is at least agreed upon that it was in full swing by the 1990's (Lee et al. 2021). New developments in ICT technology had become widespread enough to birth the term "information society" and a growing worry about the decreasing need for skilled manual labourers began to arise (Karvonen 2012). Automation as a field of technology grew into itself during this revolution, but was mostly the old, or "dumb" type of automation, machines doing a simple task like screwing a bolt or a robotic hand welding a simple seam in a factory line.

After the turn of the new millennium however, even more advances were made in making technology more autonomous and interconnected, which led to Klaus Schwab coining the term "fourth industrial revolution" in a book by the same name in 2016 (Schwab 2016). In this book, Schwab compares this industrial revolution to the first, where once fossil fuel powered machinery replaced a lot of the need for human or animal muscles, now with the fourth industrial revolution, increased computational power is replacing much of the need for human cognitive work.

Some see this as just an extension of the third industrial revolution, as much of the key technologies have their roots there. Lee et al.'s conclusion in their article comparing the differences between the third and the fourth industrial revolution was ultimately that the fourth can be considered as the second wave of the third (Lee et al. 2021). They do however acknowledge that many key technologies of the fourth industrial revolution are clearly the next evolutions of their forebearers, such as IoT to the internet, or cloud computing to personal computers, which ultimately makes the term "fourth industrial revolution" useful when discussing the impact these technologies continue to have.

2.3 Key technologies of the fourth industrial revolution

The characteristics of the fourth industrial revolution are interconnectivity of people and various systems (Schwab 2016), which has enabled among other things, increased computational power, without increasing the power of individual machines and more autonomous operations from distributed systems. In this chapter some of the key technologies that have emerged as major factors making the fourth industrial revolution possible are analysed, using the list compiled by Lee et al.'s in their 2021 article, and the technologies mentioned by Klaus Schwab in his book "The fourth industrial revolution". Other potentially significant technologies in the future include things such as the advances continually happening in the field of robotics, augmented reality and virtual reality technologies and quantum computing, but as they have not been brought up as key technologies by the primary sources used here, they will not be discussed in greater detail.

2.3.1 Internet of Things

Perhaps the most important development of the latest industrial revolution, the internet of things or IoT refers to many interconnected mechanical and/or digital devices, that can independently share information over a network (Ramgir 2019) and usually perform some complex process, be that the many devices that keep a modern home in the optimal condition for the occupant, or the multitude of machines in a factory that pick and place objects, clean, or measure conditions in a specific area to determine correct actions for optimal functionality.

The fourth industrial revolution is sometimes referred by the term "Industrial Internet of Things", which highlights just how important this development was for it. The term "Internet of Things" first appeared at the turn of the millennium, which is when the first implementations of the concept started development. By 2010 the first industrial applications for IoT were introduced and it is estimated that by 2025, there will exist more than 250 billion devices as parts of various IoT systems (Nidagundi 2022).

2.3.2 3D – printing

Additive manufacturing, or more commonly 3D-printing technologies refer to the process of manufacturing where material is added layer by layer in a desired shape to produce highly customizable parts. Developed around 40 years ago (Kalaskar 2017), conventional 3D-printing has branched out significantly from its original roots as a way to make industrial prototypes.

While the traditional plastic filament printing is still a large industry, more modern innovations have made it possible to print with materials such as metals, or organic materials. 3D bioprinting is in fact a large emerging field in biomedical research and tissue engineering (Kalaskar 2017).

2.3.3 Cloud computing

According to Marinescu, cloud computing was defined as “a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.” by the US National Institute of Standards and Technology back in 2011 (Marinescu 2017). In layman’s terms, this refers to the process of utilizing computational power from many different systems that are not necessarily tied to a location.

With cloud computing, processes that would be impossible with only the hardware at hand suddenly become a reality. Cloud computing can be used to enhance IoT systems, power entertainment systems that only require an internet access, or solve complex calculations in processes such as protein folding or large-scale data analysis.

2.3.4 Big data

Big data can broadly be defined by three key characteristics, which according to Hurwitz et al. are extremely large volumes of data, at extreme velocities and with great variety of it (Hurwitz et al. 2013). Big data in other words is the offspring of efficient data gathering and management technologies, which allow the gathering, processing and analysing of ever larger quantities of data in an efficient manner. Big data also includes the capability to perform these processes in real-time, which in turn allows for immediate reaction to developing situations.

Data types in big data can be vast and the interplay of industry 4.0 technologies can be clearly seen in its processing. The data can be collected by IoT devices, processed using cloud computing due to the sheer amount of it and analysed by AI to quickly gather insights (Lee et al. 2021).

2.3.5 Artificial intelligence

Current AI technologies mostly refer to large language models, which are an extremely new development in the public eye. Trained with a gargantuan amount of data, utilizing new developments in deep learning, natural language processing and made possible by ever increasing computational power (Amaratunga 2023), new and widely available AI have been making head waves as the next big thing in tech. As previously mentioned, AI tools are extremely efficient in analysing large quantities of data efficiently and often incredibly accurately, in a fraction of the time it would take a human. In industrial settings these tools are used in generating simple code quickly, as an aide in analysing information about specific products and as a translator between different parties (Loukides 2023).

In the future these tools may play some part in ERP systems, which would in turn give them a sort of managerial position within a company, or on the field within production facilities, where AI tools may partly aide operators in their daily tasks, and in turn enhance production capabilities. These tools will most likely only get better and more sophisticated, which is why these are just a few of the multitude of possible functions they can perform at any level of automation.

2.4 Interplay of key technologies

Interconnectivity is a vital feature of the fourth industrial revolution, and because of this, various future applications are likely to use a set of technologies to achieve truly revolutionary results.

Lu et. al display one of these possible applications in the field of medicine in their book (2021).

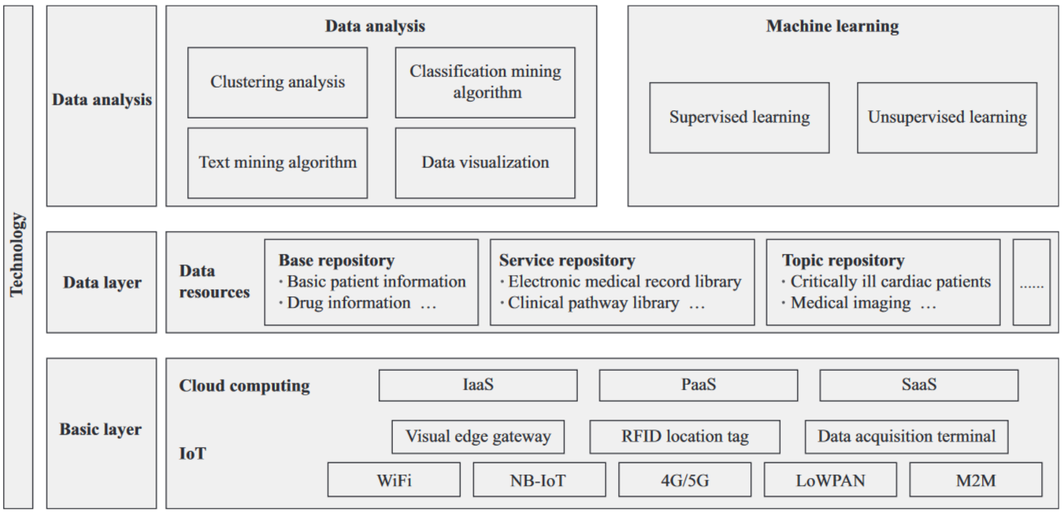

Figure 1. *Technical framework of the application of AI in clinical medicine. The different subsections within larger blocks represent the various technologies and methodologies needed to bring them to life (Lu et al. 2021, p. 1136)*

This particular application demonstrates truly what is possible when most of the technologies of the fourth industrial revolution are made to work in tandem. Through a huge amount of data, analysis can be performed using machine learning algorithms utilizing both supervised and unsupervised learning. Data is stored in cloud servers, where cloud processing takes place, using external code services developed for this purpose. Data gathering and transference is done through various IoT applications. Lu *et al.* also provide a potential service built using their framework.

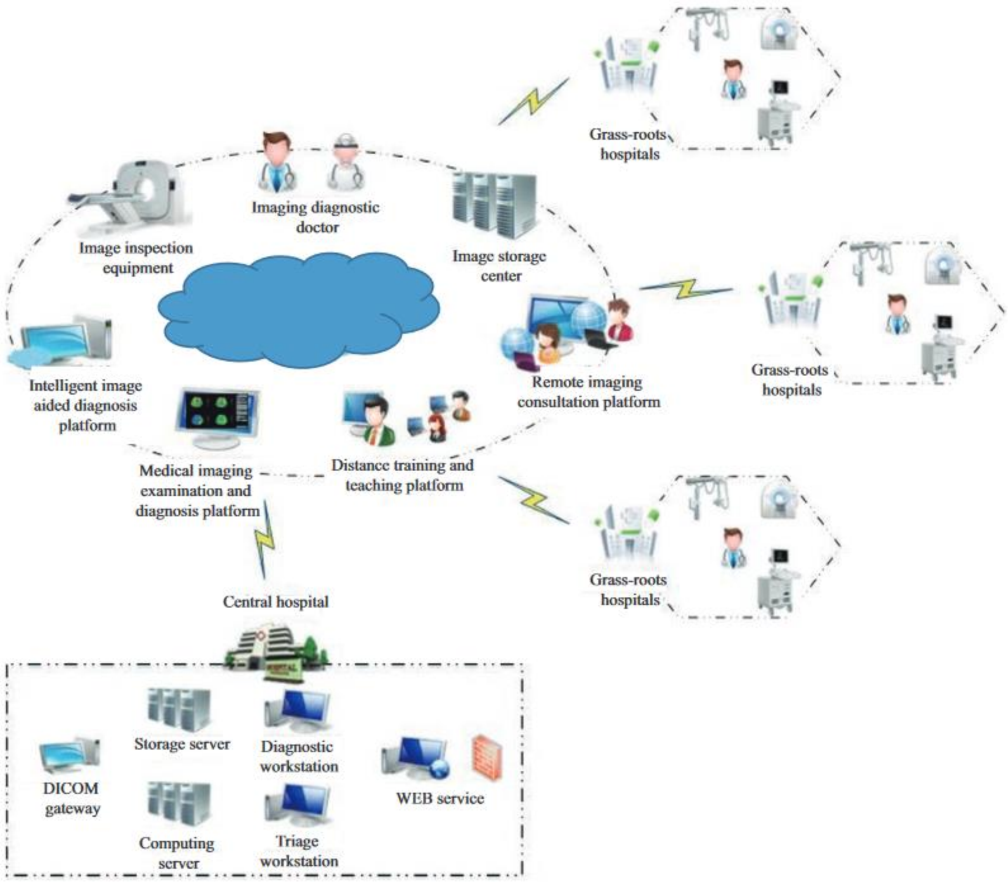

Figure 2. *Schematic diagram of a remote imaging consultation platform with a central hospital as the main body. (Lu et al. 2021, p. 1141).*

Lu et al. propose that besides working as a reactive tool to emerging problems with patients, this solution could be used as a pre-emptive measure. With enough data and more accurate measurements than a doctor may be able to gather during a traditional visit, various treatable health problems may be able to be prevented, thus increasing chances for better quality of life in some cases. These kinds of systems may also be used as a type of ERP between hospitals. Resources vary by location, be they material or human, so a well-built system should be able to allocate the necessary staff, patients and materials according to each grass roots hospital's needs and capabilities. This would eliminate unnecessary bureaucracy and save precious time in dire cases.

This is but one example of what the key technologies of the fourth industrial revolution may bring with them. If executed and implemented properly, this kind of application may free up valuable time from doctors and nurses, by eliminating the more cumbersome systems and processes used today. Humans are prone to errors and delays, and while machine learning and LLM applications should always be fact checked for their contribution's validity, and complex "smart" systems continually developed and evaluated, they are undoubtedly faster in retrieving and processing data than a live human, which in turn is immeasurably valuable in certain areas.

3. COMPARING INDUSTRIAL REVOLUTIONS

3.1 Introduction to the methodology

In this chapter, the impact of different innovations during various industrial revolutions is analysed. The aim is to find out common factors between the effects key innovations have had during previous industrial revolutions. After this, the goal is to examine the current fourth industrial revolution and whether the technologies described in the previous chapter have had a comparable impact. Since the fourth industrial revolution is very much ongoing, the expected problem will be comparing past events by their impact to current ones, since the full-scale picture is yet to be formed.

3.2 Selecting data

Due to major historical events, GDP growth is not observed when delving into the effects of industrial revolutions. A better metric to use would be the productivity of labour, but as established by Nicholas Craft in his paper about the slowness of economic growth compared to the rapid rate of industrialization during the first industrial revolution, there is a delayed effect between the adoption of new technologies and productivity of labour (Craft 2005). The data on productivity of labour from the first industrial revolution is also scarce, the Encyclopedia Britannica puts the growth in productivity from 1760 to around 1810 at around 0,5% per year, after which it grew to around 1% per year (Frankel et al. 2024). From the following table the growth of productivity seems to rise until the third industrial revolution, but growth slows down significantly during the beginnings of the third revolution.

		1870–1913	1913–50	1950–73	1973–84
United States		2.0	2.4	2.5	1.0
five-country average		1.6	1.6	5.3	2.8
the five countries	France	1.7	2.0	5.1	3.4
	Germany	1.9	1.0	6.0	3.0
	Japan	1.8	1.7	7.7	3.2
	Netherlands	1.2	1.7	4.4	1.9
	United Kingdom	1.2	1.6	3.2	2.4

Figure 3. *Phases of growth in labour productivity from 1880 to 1984. The comparison is done between western countries which adopted the technologies of the second industrial revolution at a similar time (Maddison 1987).*

When looking at the data about hours worked, it seems to support this. Productivity of labour essentially just means hours worked per unit produced, in which case the average working hours give us an insight into how technology had an effect on industry. The following graph details data gathered from various sources, displaying the average working hours.

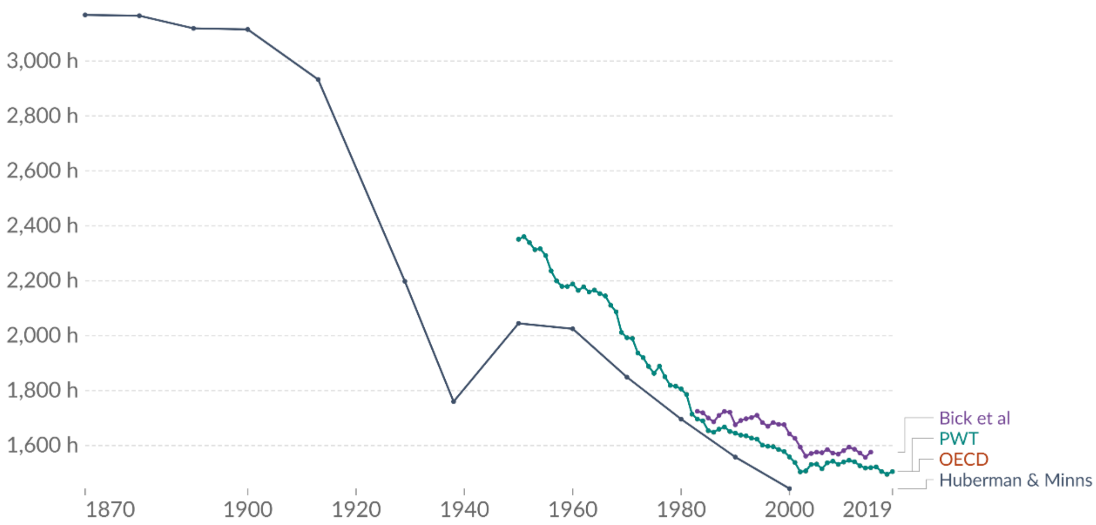

Figure 4. *Annual working hours per worker, various sources, France. While only one of the four sources on this graph displays pre-1900's data, the trends of the 20th century seem to match what later research has found (Giattino et al. 2020).*

Again, the second industrial revolution appears to have had an enormous effect, after which (omitting the obvious gap caused by the second world war here), lessening of working hours seems to diminish.

Thus, the effects of the fourth industrial revolution cannot at the present moment be observed through these statistics, and the observations have to focus on how industrial settings and other places of work have changed with the adoption of new technologies.

3.3 The effect of key innovations during industrial revolutions

The first comparison will be between the first and the second industrial revolutions. As previously established, the notable developments of the second industrial revolution were the widespread adoption of electricity, the internal combustion engine, advances in chemistry that brought us things like petroleum, and the first wireless communication technologies that spawned the radio. In the following graph from Atkeson et al.'s study on the second industrial revolution, the change in mechanical drives within manufacturing plants can be seen.

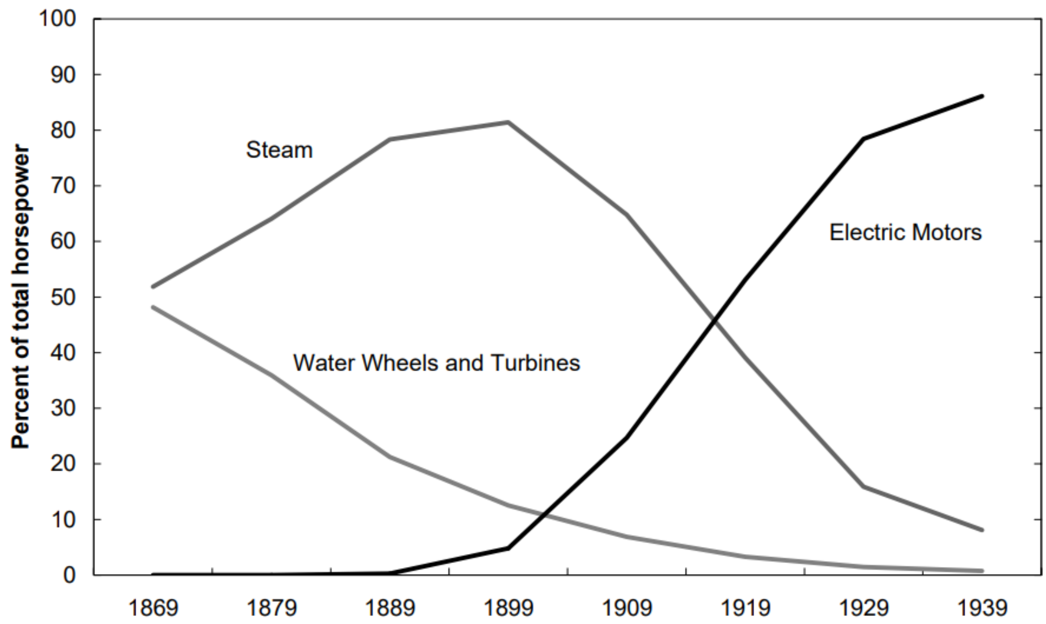

Figure 5. *Sources of Mechanical drive in U.S manufacturing establishments from 1839-1939. This graph shows the transitionary period when the technologies of the second industrial revolution became more widely adopted (Atkeson et al. 2001).*

As electrical motors got better, their adoption rates skyrocketed, and the manufacturing plants of the first industrial revolution slowly replaced their steam and water powered engines with electrical ones. The latter half of the second industrial revolution also saw the rise of mass production, undoubtedly helped by the widespread implementation of more efficient electrical motors and the advances in communications and transporting technologies (i.e. trains, boats and cars powered by an internal combustion engine rather than steam).

The transition from the second to the third saw the transition from analogue format to digital, with inventions such as the transistor making computers viable to be used in almost any setting. This made it possible to integrate them into all aspects of industry, which in turn opened up the door for these computers to be linked together to form first closed networks and later what we now know as the internet. This is why the third industrial revolution is sometimes referred to as the digital revolution. Digitalized devices became a key aspect of everyday life during this revolution, which can be seen in the following graph depicting the number of computers in use worldwide.

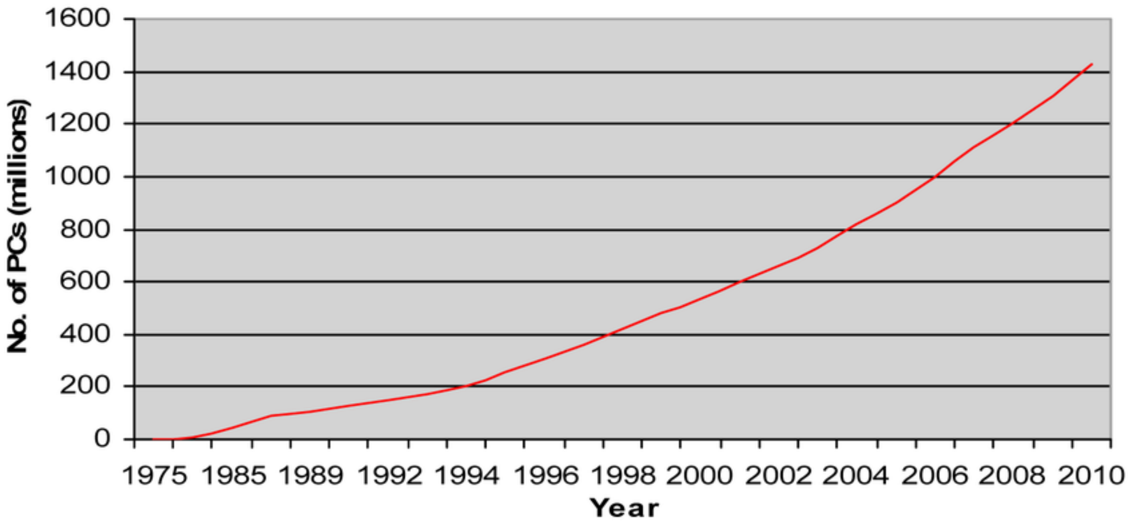

Figure 6. *The number of computers, including carriables, in use worldwide over time (Somavat et al. 2011).*

Somavat et al.’s research on the topic (2011) failed to see the explosive raise of handhelds in the 2010’s, but later research shows just how widespread they eventually became, in places overtaking PC’s as the number one form of communication technology. From the following graph, the adoption of selected communication technologies during the whole of the third industrial revolution can be observed.

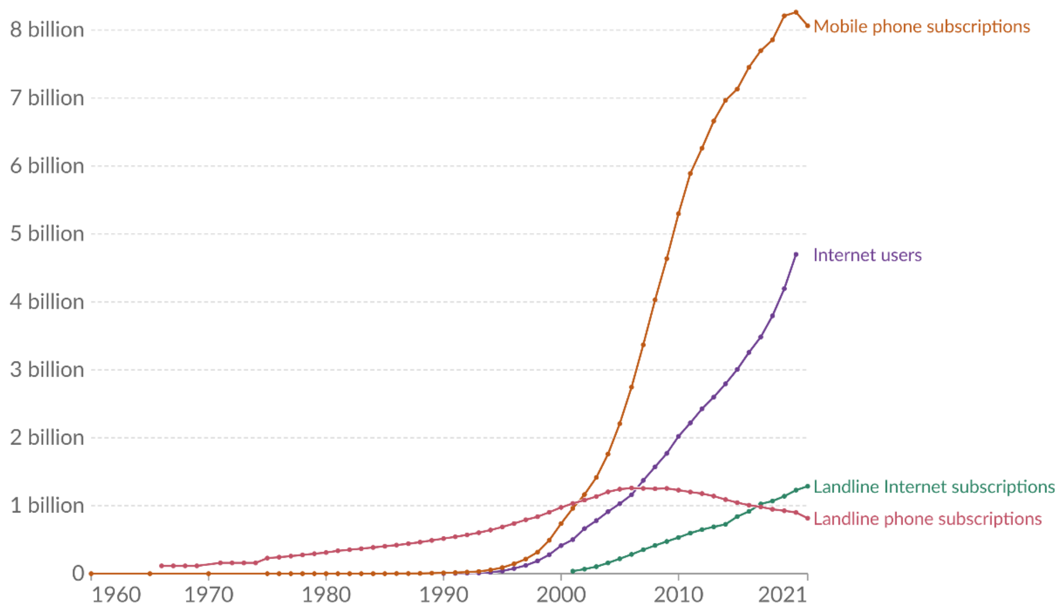

Figure 7. *The adoption of communication technologies over time. With wireless communication technologies rising rapidly in popularity during the 21st century, traditional physical connections are falling out of fashion (Roser et al. 2023).*

Internet adoption began its meteoric rise at the height of the third industrial revolution, around the 1990's, and as can be seen from figure 7. Along with mobile phone subscriptions, they are the overwhelmingly dominant form of modern connectivity. The data in figure 7 is somewhat incomplete, and a negative trend cannot be established from a one-year downshift in mobile phone subscription popularity, but the amount of internet users will most likely at some point at least catch up to the amount of mobile phone subscriptions. One thing to note is that as the number of mobile phone subscriptions is around the same as the number of people alive at the moment, it most likely is slightly skewed from the expected one subscription per person, due to the fact that a large number of people have more than one subscription for things such as work.

The fourth industrial revolution is characterized as a rise in interconnectivity of devices, large range of customization of parts and large amounts of data, processed over an internet connection and analysed by an AI. The various technologies of the fourth industrial revolution have been experiencing rapid development, as can clearly be seen in Lee et al.'s article (2021), where they detail the number of patents each of key technology.

4IR technologies	2010	2011	2012	2013	2014	2015	2016	2017	2018	Annual avg. growth
3D printing	4	2	5	8	12	34	61	110	188	78%
AI	29	26	38	43	37	41	44	44	78	16%
IoT	4	4	10	13	5	20	37	116	276	107%
Big data	5	5	6	9	10	32	46	81	106	56%
Cloud computing	3	15	41	166	320	394	437	425	428	125%
4IR Total	45	52	100	239	384	521	625	776	1076	53%
4IR average	9.0	10.4	20.0	47.8	76.8	104.2	125.0	155.2	215.2	53%
Share (=4IR/Total)	0.020%	0.023%	0.039%	0.085%	0.127%	0.173%	0.208%	0.257%	0.369%	47%
Total no. of US patents	220,719	225,777	254,668	279,900	303,450	301,325	300,091	302,379	291,674	4%

3IR technologies	1990	1991	1992	1993	1994	1995	1996	1997	1998	Annual avg. growth
Mobile phone	3	3	1	11	6	24	22	45	51	162%
ASIC	2	3	7	11	14	25	33	35	46	52%
Internet			1	2	4	4	6	26	171	190%
Memory chips 1	83	100	98	125	150	172	227	214	356	22%
PC1	69	80	77	130	147	177	211	256	395	26%
Memory chips 2(76~84)	40	31	43	45	33	40	43	51	58	7%
PC2(82~90)	2	1	3	8	16	33	31	69	69	80%
3IR total 1	157	186	184	279	321	402	499	576	1019	28%
3IR total 2	47	38	55	77	73	126	135	226	395	35%
4 3IR total 3 (= mobile phone + ASIC + Internet + PC2)	7	7	12	32	40	86	92	175	337	71%
3IR average 1	31.4	37.2	36.8	55.8	64.2	80.4	99.8	115.2	203.8	28%
3IR average 2	9.4	7.6	11.0	15.4	14.6	25.2	27.0	45.2	79.0	35%
4 3IR average 3	1.8	1.8	3.0	8.0	10.0	21.5	23.0	43.8	84.3	71%
Share1 (=3IR total 1/total)	0.173%	0.192%	0.188%	0.282%	0.313%	0.393%	0.451%	0.507%	0.675%	19%
Share2 (=3IR total 2/total)	0.052%	0.039%	0.056%	0.078%	0.071%	0.123%	0.122%	0.199%	0.262%	27%
Share3 (=3IR total 3/total)	0.008%	0.007%	0.012%	0.032%	0.039%	0.084%	0.083%	0.154%	0.223%	62%
Total patents by year	90,592	96,909	98,050	98,960	102,513	102,299	110,762	113,642	150,959	7%

Figure 8. *Patents at the of the third and fourth industrial revolutions. The timeframes for these patents are chosen to represent similar periods from both of these revolutions. (Lee et al. 2021).*

As we are currently still in the throes of the fourth industrial revolution, and as previously established, the diffusion of technology and the growth following it only happen sometime after the start of the revolution (Crafts 2005), it is somewhat impossible to find relevant data about how industrial settings have evolved. This table does however clearly show that ever increasing interest in the development of the technologies of the fourth industrial revolution is present and while the dataset is a bit dated when discussing ongoing events, it can within reason be speculated that the interest in these key technologies will only continue to increase.

4. ANALYSIS

4.1 Analysing data from different industrial revolutions

What becomes clear when looking at the data presented, is that there is no simple answer to what makes an industrial revolution. Certain sources disagree on the number of revolutions, there are no clear absolute metrics that indicate the start and end of an industrial revolution and the term itself seems more like a common agreement, since all four of the revolutions discussed here have widely different characteristics. The only observable trend seems to be that industrial revolutions begin every 100 or so years, but the fourth industrial revolution does not follow this, and it is more of a curious small-scale pattern, rather than an absolute rule.

4.2 The impact of the technologies of the fourth industrial revolution to the nature of work

The technologies of the fourth industrial revolution have steadily gained popularity in a variety of industries in the past decade. Their impact however remains somewhat inconsistent. Focusing a lens on how they affect the way on which people work gives us some insight, however.

Cloud computing using serverless cloud architecture is steadily replacing data centres as the standard form of storing one’s data and application functions (Marinescu 2022). Services such as Azure, AWS and Google Cloud allow their users to access much more processing power at a much cheaper price than what one could find in a traditional data centre. Because of the relative cheapness of utilizing these cloud providers, developing new systems is no longer hindered by processing power, or storage space, but rather an internet connection. This does not directly revolutionize how different digital services are utilized and developed, but it does require a different set of skills when compared to developing against a physical server and it possibly reduces the need for technicians, since physical installations are no longer needed as frequently.

Internet of things applications have seeped into the consumer market more and more, and in industrial settings, can be seen in things such as fleets of self-driving pallet movers, sensors which detect various errors and inconsistencies and allowing an operator to get real time data on where their attention is needed. Their work thus has shifted more towards monitoring the conditions of the industrial setting and the work done by automated systems.

Additive manufacturing or 3D-printing currently allows much more customization in which parts are produced to which purpose. This has a two-fold effect; on one hand every part of a machine can be reproduced to replace wear and tear, which eliminates the need to store these parts and brings the price of more niche parts down and on the other hand it will most likely have a negative effect on standardization, due to manufacturers not needing to rely on providers of these parts to have their specific need in stock. Because of this, there is no need to conform to standardization as a part of the design process. Concrete ways how this has affected on how people work is yet to be seen.

Big data and artificial intelligence go somewhat hand in hand, since the former is used to train and optimize the latter. Large language models are extremely good in searching through large amounts of data and summarising key findings, which is why work that requires memorization, while already in decline due to the internet and various search engines, is quickly becoming more and more obsolete.

4.3 Labour and the fourth industrial revolution

As can be extrapolated from figures 2 and 3, the amount of labour a person is subjected has been on a downward trend since the first industrial revolution. This is especially true for manual labour, with the adoption of different types of mechanical apparatuses used in manufacturing. The third and fourth industrial revolutions however have had an even larger effect on intellectual work than the previous ones and thus it has in fact increased as more and more people focus on more skilled labour. This is further supported by the report “The Changing nature of work” by the World Bank (2019). The report shows that countries, that have traditionally adopted new industrial revolutions earlier are losing their industrial workforce, while countries in east Asia are increasing theirs.

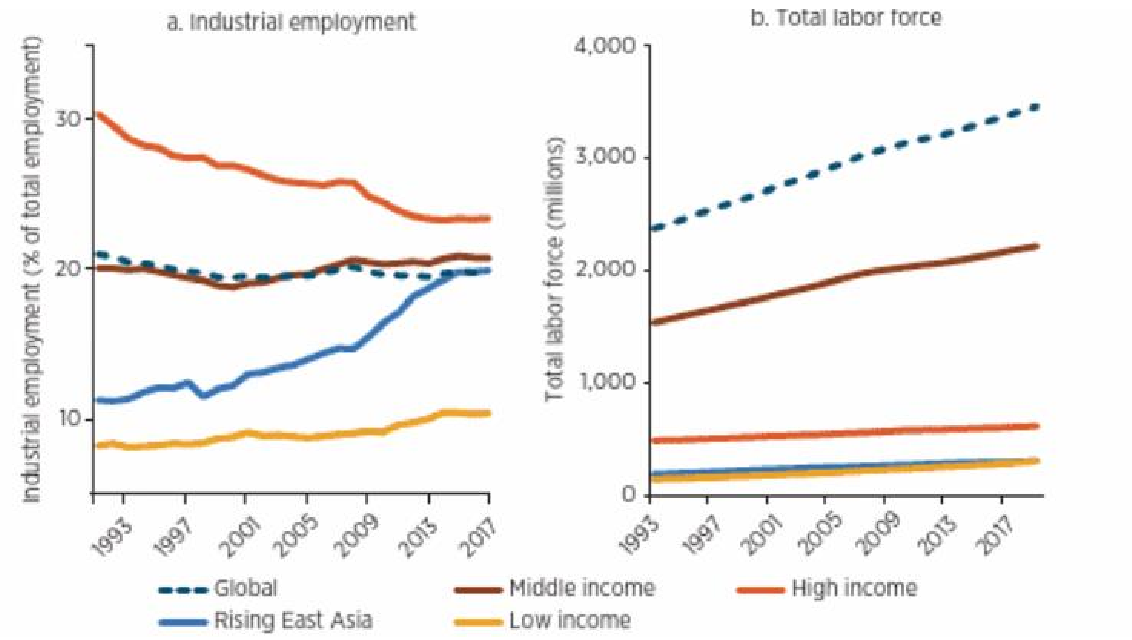

Figure 9. *While the traditionally wealthy countries are experiencing loss in industrial jobs, countries which industrialised later and are rising economically are experiencing the opposite. The world's total labour force is simultaneously growing (The World Bank 2019).*

The loss of industrial jobs and the simultaneous rise of total workforce essentially means, that people are transitioning to more high skilled work. This is not without its problems however and the moniker of "skilled work" is alas a bit misleading, as ever more complex systems require extremely specific knowledge to operate or are often made inefficient by their complexity and often include a lot of processes, which sink up valuable time from highly educated workers. Whether it be doctors spending most of their working hours inputting patient information, or automation operators manually resetting machinery after minor abnormalities, which would not in reality require any action, these processes could very well be automated in the future by the technologies of the fourth industrial revolution, and thus leaving human workers to higher decision making and strategizing.

With these changes, the effect this fourth industrial revolution will have, might actually be quite similar to the previous ones, when it comes to ways of working. If traditionally white-collar professions can be said to have a manual component, such as writing and fidgeting with various systems, then this industrial revolution might follow the previous ones in reducing the amount of manual labour a worker is subjected to, although in fields where this is not traditionally considered.

Whether this can be considered to change the nature of work on par with the other revolutions remains to be seen, but an argument can certainly be made that the changes at the very least seem potentially significant. As with all industrial revolutions, technologies are not adopted uniformly worldwide, which leads to variance based on geographical location in the nature of work.

5. CONCLUSION

The different technologies presented here and compared to the key technologies of the previous industrial revolutions ultimately cannot be said to have had a comparable effect so far. This is mainly due to the fact that we are currently living amidst the fourth industrial revolution and cannot yet know the full-scale impact it will have. Klaus Schwab only announced the fourth industrial revolution to have begun in 2016, and since the previous ones are thought to have lasted for more than 50 years, we are in an impossible position to judge the effects as of now. Nevertheless, if key technologies to describe the fourth industrial revolution would have to be put forth, the internet of things, 3D-printing, cloud computing, big data and AI are the clear frontrunners, based on existing literature and research. The key technologies of the previous revolutions have had a clear connection to one another, such as with the second revolution's petroleum and internal combustion engine, or the third's computers and the internet. With the fourth, each technology has a clear way of playing off one another, to a great extent. Since connectivity of various key resources made possible by these technologies is the defining characteristic of this revolution, the name sometimes used for it, industrial internet of things, is more than applicable.

The definition of an industrial revolution remains somewhat vague, but with the previous three having reduced and altered what kind of work is being done and how much the average worker both works and produces with their effort, it can be used as the main comparison point between industrial revolutions, as well as the effect one would look for when trying to define an industrial revolution. While we are still in the throes of what could be the fourth industrial revolution and cannot clearly see the end results of this era, the key technologies listed here certainly have the potential to produce this, even though the steps to get there cannot be known for certain as of yet.

The broad technologies listed here, and their future applications may not have yet kickstarted a new way of working for the majority of the human populace, but the type of work being done is changing at a varied pace around the whole world, with the effects one would assume the fourth industrial revolution would bring being most strongly seen in more developed countries with a highly educated workforce, in the near future.

SOURCES

Anon (2019) World Development Report 2019. The Changing Nature of Work. Washington, D.C: The World Bank.

Amaratunga, T. (2023). Understanding Large Language Models Learning Their Underlying Concepts and Technologies. Apress. Available: https://learning.oreilly.com/library/view/understanding-large-language/9798868800177/?sso_link=yes&sso_link_from=tampere-university.

Atkeson, A., Patrick K. (2001). The Transition to a New Economy After the Second Industrial Revolution. Mass: National Bureau of Economic Research. p. 44. Available: https://www.nber.org/system/files/working_papers/w8676/w8676.pdf

Crafts, N. (2005). The First Industrial Revolution: Resolving the Slow Growth/Rapid Industrialization Paradox. Journal of the European Economic Association. p. 525–534. Available: https://openurl-ebsco-com.libproxy.tuni.fi/linksvc/linking.aspx?sid=Primo&volume=3&date=20050401&spage=525&issn=1542-4766&issue=2%2F3&genre=article&title=Journal+of+the+European+Economic+Association.&epage=534&doi=10.1162%2F1542476054472955

Giattino, C., Ortiz-Ospina, E., Roser, M. (2020). Working Hours. OurWorldInData. Available: https://ourworldindata.org/working-hours

Frankel, M., Kendrick, J. (2024). Historical trends: Early industrialization. Encyclopedia Britannica. visited 6.1.2024. Available: https://www.britannica.com/money/topic/productivity/Historical-trends

Hounshell, D. (1985). From the American System to Mass Production, 1800-1932 the Development of Manufacturing Technology in the United States. Baltimore: Johns Hopkins University Press. p. 1.

Hudson, P. (2014). The Industrial Revolution. Hodder Arnold. p. 2-5. Available: https://ebookcentral.proquest.com/lib/tampere/reader.action?docID=1778888

Hurwitz, J. (2013). Big Data for Dummies. Ind: John Wiley & Sons. p. 48-58.

Kalaskar, D. (2017). 3D Printing in Medicine. Elsevier. p. 2-8.

Karvonen, E. (2012). Informational Societies : Understanding the Third Industrial Revolution. Tampere University Press. p. 9-20. Available: https://trepo.tuni.fi/handle/10024/65316

Lee, J. & Lee, K. (2021). Is the Fourth Industrial Revolution a Continuation of the Third Industrial Revolution or Something New under the Sun? Analyzing Technological Regimes Using US Patent Data. Industrial and corporate change. p. 137–159. Web.

Loukides, M. (2023). What Are ChatGPT and Its Friends? : Opportunities, Costs, and Risks for Large Language Models. CA: O'Reilly Media, Inc. Available: https://learning.oreilly.com/library/view/what-are-chatgpt/9781098152604/?sso_link=yes&sso_link_from=tampere-university

Lu, Zx., Qian, P., Bi, D. et al. (2021). Application of AI and IoT in Clinical Medicine: Summary and Challenges. CURR MED SCI 41, 1134–1150. Available: https://doi-org.libproxy.tuni.fi/10.1007/s11596-021-2486-z

Maddison, A. (1987) Growth and Slowdown in Advanced Capitalist Economies: Techniques of Quantitative Assessment. Journal of economic literature. 25 (2), 649–698.

Marinescu, D. (2022). Cloud Computing: Theory and practice. Elsevier Science & Technology. p. 1-18. Available: https://ebookcentral.proquest.com/lib/tampere/detail.action?pq-origsite=primo&docID=1213925

Nidagundi, P. (2022). The IoT Product Manager: A Handbook for Engineers, Data Analysts, and Other IT Professionals. Apress L. P. p. 1–18. Available: https://learning.oreilly.com/library/view/the-iot-product/9781484286319/?sso_link=yes&sso_link_from=tampere-university

Petrosyan, A. (2023). Number of internet and social media users worldwide as of October 2023. Statista.com, visited 15:16 21.12.2023, Available: https://www.statista.com/statistics/617136/digital-population-worldwide/.

Ramgir, M. (2019). Internet of Things. Pearson Education India. Available: https://learning.oreilly.com/library/view/internet-of-things/9789353941529/?sso_link=yes&sso_link_from=tampere-university

Roser, M., Ritchie, H., Mathieu, E. (2023). Technological Change. OurWorldInData. Available: https://ourworldindata.org/technological-change

Schwab, K. (2017). The Fourth Industrial Revolution. London: Portfolio Penguin.

Somavat, P. & Namboodiri, V. (2011) Energy consumption of personal computing including portable communication devices. Journal of Green Engineering. 1 (4), 447–475.

Vincent, J. (2016). UN condemns internet access disruption as a human rights violation. theverge.com. Available: https://www.theverge.com/2016/7/4/12092740/un-resolution-condemns-disrupting-internet-access